

AI-Assisted Mixed-Signal Physical Design

Raj Joshi

Week 1 Task Report – PLL IP Design Track

This report summarises a Week 1 study of the `nitjsr_pll_130nm` repository, which implements an $8\times$ clock-multiplier PLL in SKY130 130 nm technology using open-source tools (`xschem`, `ngspice`, `Magic`). AI tools were used to decompose the problem, generate Codex-style prompts, and structure the study into executable research blocks. Every result documented here was understood and verified by the author; no AI output was accepted without independent scrutiny against repo files, SKY130 model cards, and tool documentation.

I. OBJECTIVE

The objective was to understand the fundamentals of Phase-Locked Loop (PLL) design and to study the reference repository `himansh107/nitjsr_pll_130nm` without reproducing full simulation results in Week 1. The work was divided into: repo understanding, PLL block decomposition, SKY130/`xschem/ngspice` tool-setup study, pre-layout versus post-layout simulation comparison, probable failure modes, and documentation.

Special focus was placed on understanding how each sub-block—Phase Frequency Detector (PFD), Charge Pump (CP), Loop Filter (LF), Voltage Controlled Oscillator (VCO), and Frequency Divider ($\div 8$)—interacts within the feedback loop to achieve frequency locking.

II. REFERENCE REPO SUMMARY

The repo implements an $8\times$ clock multiplier targeting a **500 MHz output** from a **62.5 MHz reference** input using SKY130 PDK primitives. The five major blocks are: (1) **PFD**: generates UP/DOWN pulses from the phase/frequency difference between reference and feedback clocks; (2) **Charge Pump**: drives current into or out of the loop filter based on the UP/DOWN state; (3) **Loop Filter**: a passive RC network converting CP current to a smooth control voltage V_{ctrl} ; (4) **VCO**: a current-starved ring oscillator whose frequency is proportional to V_{ctrl} ; (5) **Divide-by-8**: three cascaded toggle flip-flops feeding the divided VCO output back to the PFD. Pre-layout netlists are simulated directly in `ngspice`; post-layout simulations use parasitic-extracted SPICE exported from `Magic`.

III. AI WORKFLOW

An AI assistant was used to: summarise the repo README and convert it into six executable research tasks; generate Codex-style step-by-step prompts for each PLL sub-block; produce `ngspice` plot commands and `Magic` layout-viewing steps; and identify probable setup failure points with suggested fixes. AI suggestions were treated as first-draft scaffolding and cross-checked against the actual repo source files, the SKY130 model card, and `xschem/ngspice` documentation before being documented here.

IV. PLL BLOCK STUDY

A. PFD UP/DOWN Signal Generation

C. VCO Tuning and Control Voltage

The ring-oscillator VCO uses current-starved inverters whose tail current is controlled by V_{ctrl} . The tuning gain K_{VCO} must be characterised across the expected V_{ctrl} range (0.6–1.6 V at 1.8 V supply for SKY130 tt corner). A `.dc` sweep in `ngspice` validates the frequency-vs-voltage tuning curve before closed-loop simulation.

D. Divide-by-8 Feedback Path

Three cascaded toggle flip-flops (each $\div 2$) implement the $\div 8$ counter. Propagation delay through the chain adds phase lag in the feedback path and must be budgeted in loop stability analysis. At lock, $f_{\text{FB}} = f_{\text{VCO}}/8$ must equal $f_{\text{REF}} = 62.5$ MHz.

V. RESULTS AND OBSERVATIONS

The study was completed at the conceptual and structural level. Key observations: (1) SKY130 model files must be referenced via a `.lib` card pointing to the correct corner (tt, ff, ss) from the `open_pdk` installation; (2) `ngspice` `.tran` timestep must be ≤ 2 ps to resolve 500 MHz signals without aliasing; (3) post-layout extraction in `Magic` (`extract all` $\rightarrow$ `ext2spice`) produces an RC-parasitic netlist that replaces the schematic netlist in `ngspice` for post-layout lock-time estimates; (4) expected lock time from repo waveforms is approximately 3–5 μs with the stated loop filter values.

VI. CHALLENGES AND DEBUGGING

Likely failure modes and suggested fixes: (1) **Missing `.lib` path**—`ngspice` exits with “model not found”; fix by verifying the absolute path to `sky130.lib.spice`; (2) **VCO not oscillating**—check V_{ctrl} is within the valid tuning range; reduce ring stage count if free-running frequency is too low; (3) **PFD stuck**—probe `v(up)` and `v(dn)`; AND-reset must de-assert within ≈ 10 ps; (4) **Magic DRC errors after extraction**—correct geometry, then re-run `extract all` and `ext2spice`. Next steps include running the pre-layout `ngspice` simulation and capturing a V_{ctrl} -vs-time transient to validate PLL locking behaviour.

[1] H. Verma, “nitjsr_pll_130nm,” GitHub, 2023. Available: github.com/himansh107/nitjsr_pll_130nm

[2] Google/SkyWater, “SKY130 PDK Documentation,” skywater-pdk.readthedocs.io, 2020.

The PFD uses two D-flip-flops clocked by REF and FB respectively, with an AND-reset path. When REF leads FB, the UP signal goes high; when FB leads REF, DOWN goes high. A dead-zone capacitor at the reset node prevents simultaneous UP+DOWN pulses that would create a blind spot near lock. Verifying this requires a `.tran` simulation probing `v(up)` and `v(dn)` with clocks at slightly different frequencies.

B. Charge Pump and Loop Filter

The CP consists of a PMOS current source switched by $U\bar{P}$ and an NMOS current sink switched by DOWN. Mismatch between source and sink currents introduces a static phase error at lock. The loop filter ($R \approx 200 \text{ k}\Omega$, $C \approx 500 \text{ fF}$) sets the loop bandwidth and damping factor ζ ; a larger capacitor improves phase margin but increases lock time.

- [3] R. J. Baker, *CMOS Circuit Design, Layout, and Simulation*, 3rd ed., IEEE Press/Wiley, 2010.